

Nabil Rakaiza Abror

nabilraka1234@gmail.com | +65 8917 6960 | github.com/nabilrakaiza | linkedin.com/in/nabilrakaiza | nabilrakaiza.vercel.app

EDUCATION

National University of Singapore

Aug 2024 – Jun 2028

B.Sc. (Hons.) Data Science and Analytics, Second Major in Computer Science

GPA: 4.63/5.00 (Highest Distinction) | Honors: Top Student — CS1010S, IT1244

EXPERIENCE

Distribution Operations Intern, Singlife, Singapore

May 2026 – Aug 2026

- Automated a monthly cross-partner Qlik + Python reporting pipeline, cutting a full 6-partner report run from ~90 to ~18–20 minutes (80%) and new-partner onboarding from ~30 to ~1.5 minutes
- Built 5+ Qlik apps consolidating siloed Excel-based reports into unified, scheduled pipelines feeding stakeholder decisions
- Wrote a Python automation script to apply Excel sensitivity labels and package AES-encrypted deliverables for automated, secure report distribution

Data Science Intern, Quantum Teknologi Nusantara, Indonesia

May 2025 – Jul 2025

- Built and deployed 3 AI systems using Python and LLM frameworks, including a CVRPTW route optimizer, an AI graphic designer, and an ontology-based news relationship engine
- Improved LLM output relevance and consistency by ~20% through prompt decomposition, structured outputs, and evaluation loops

Undergraduate Teaching Assistant (CS2030, CS2040), NUS

Jan 2026 – Present

- Selected to TA two consecutive CS modules; mentored 20+ students weekly in object-oriented design and data structures, debugging and reviewing student codebases

PROJECTS

Indo-English Translation Machine | [GitHub Link](#) | [Live Demo](#)

- Built a Transformer-based Indonesian→English NMT model from scratch in PyTorch with a custom BPE tokenizer, trained on Helsinki-NLP/opus-100 (Colab T4)
- Deployed as a live Gradio app on Hugging Face Spaces, integrated into personal portfolio site

Automatic Survey Analyzer | [GitHub Link](#)

- Led development, as primary developer on a 3-person team, of an end-to-end survey-analysis pipeline — covering survey-design diagnostics, an XGBoost+SHAP model for incomplete-response drivers, and LLM+DBSCAN-based respondent segmentation
- Built a Streamlit interface and integrated the Gemini API for LLM-assisted question filtering and automated insight-report synthesis

Papper — Full-Stack POS System | [GitHub Link](#)

- Built a cross-platform POS system (Android + web) for a family restaurant using React Native (Expo), Supabase, and PostgreSQL, supporting dual roles with real-time order syncing
- Implemented automated stock deduction, role-based auth, and Bluetooth thermal receipt printing; eliminated a yearly SaaS subscription and now handles 50+ daily orders in production

social-sim-rag — RAG-Powered Narrative Simulation | [GitHub Link](#) | [Live Demo](#)

- Built an end-to-end RAG pipeline (chunking, 768-dim embeddings, pgvector storage, hybrid retrieval, grounded generation) giving 3 characters siloed per-playthrough memory across a 30-day simulation, with no scripted dialogue
- Built a 23-case labelled evaluation harness to raise retrieval MRR from 0.47 to 0.86, and replaced threshold-based retrieval gating with a lexical intent gate after finding greetings/questions weren't separable by similarity — taking correct no-retrieval from 0% to 100% with zero hit-rate loss

Monitoring Human Mobility with Smartphones | [GitHub Link](#)

- Designed and trained a CNN-LSTM architecture (312K params) for 12-class human activity recognition from raw smartphone IMU sequences, achieving 94.2% accuracy and 0.87 macro F1 with subject-wise train/test splitting
- Found raw sequences outperformed hand-engineered features by a wide margin on the same architecture (0.87 vs. 0.54 macro F1) — representation mattered more than model complexity

SKILLS

Programming: Python, R, SQL/PostgreSQL, Java, JavaScript, TypeScript, C++

ML & Data: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, LangChain, LangGraph, Gemini, OpenAI

Frameworks: React, React Native, Node.js, Express, Next.js, Flask

Tools: Power BI, Tableau, Qlik, Excel, GitHub, Figma, Supabase, Power Automate

Languages: English (Fluent), Indonesian (Native)